Bayesian Model Averaging and Jointness Measures for gretl – 2.1

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Abstract

Bayesian model averaging is a model-building strategy that takes account of model uncertainty in conclusions about estimated parameters. It is an efficient tool for discovering the most probable models and obtaining estimates of their posterior characteristics. We introduce the **BMA** package for linear regression models with jointness measures proposed by Ley and Steel (2007) and Doppelhofer and Weeks (2009). More information one can find in Błażejowski and Kwiatkowski (2015).

Keywords: Bayesian model averaging, jointness measures, **gretl**, Hansl.

JEL: C11, C51, C87, O47.

Changelog

2.1 Small code refreshing:

- Some code cleanups.
- Getting rid of calls to deprecated functions.
- Correct handling of missing values.

2.0 Bundle based version (most of internal functions call only `bundle M`).

1.0 Initial version.

1 Introduction

We know from elementary statistical theory that linear regression attempts to model the relationship between two or more variables by fitting a simple linear equation to observed data. In the classical approach, we usually rely on ordinary least squares (OLS) or maximum likelihood (ML) estimates and popular model selection criteria, i.e., the Akaike information criterion (AIC) and the Bayesian information criterion (BIC), to find the "best" model. A problem with this approach arises when we have to select a "good" subset of variables from a large set of regressors. When the number of possible exogenous variables is K, the number of

possible linear models is 2^K. If we have, for example, $K = 30$ possible regressors, the number of possible combinations of variables to be included or not equals 1073741824. This means that it is very difficult, if not impossible, to find the estimates for all combinations. Moreover, Raftery, Madigan, and Hoeting (1997) show that standard variable selection procedures lead to different estimates and conflicting conclusions about the main questions of interest.

Bayesian model averaging is a useful alternative to other variable selection procedures, because it incorporates model uncertainty into conclusions about the estimated parameters. Bayesian model averaging is a standard Bayesian solution to model uncertainty, where the inference on parameters is based on a weighted average over all possible models under consideration, rather than on one single regression model. These weights are the Bayesian posterior probabilities of the individual models.

There is a recent and growing literature on Bayesian model averaging. Examples of applications of Bayesian model averaging can be found in a number of works (see, for example, Hoeting, Madigan, Raftery, and Volinsky 1999, Steel 2011 for a recent overview). Our software package for parameter estimation and model comparison of linear regression models is based on Fernández, Ley, and Steel (2001a,b) and Koop (2003). We use the Markov chain Monte Carlo model composition (MC^3) sampling algorithm developed by Madigan, York, and Allard (1995) to select a representative subset of models.

Doppelhofer and Weeks (2005, 2009) define a jointness measure of dependence among explanatory variables that appear in the linear regression models. We use this measure to identify whether two variables are substitutes, complements or neither. A similar jointness measure was also proposed by Ley and Steel (2007).

In this paper, we propose Bayesian model averaging package for **gretl**, (**gretl** Team, 2015) named **BMA**. We can list several reasons why, in our opinion, it is important to address this topic. **gretl** is an increasingly popular, free, open-source software for econometric analysis, both for students and academics. It is written in pure C "by econometricians for econometricians" with the international community gathered around two men: Allin Cottrell and Riccardo "Jack" Lucchetti. Since 2009 the community has been meeting every two years at the **gretl** conference. The last conference was held in Berlin on June 12–13, 2015. **gretl** is licensed under the GPL-3 license (`http://www.gnu.org/copyleft/gpl.html`) and has been reviewed several times in various international journals. The best known reviews were written by Baiocchi and Distaso (2003), Yalta and Yalta (2007) and Rosenblad (2008). Some brief information about **gretl** was also presented in Lucchetti (2011) and Adkins (2011). Unlike most other statistical software packages **gretl** has an easy-to-use graphical user interface (GUI). Our software package is, therefore, a free and easy-to-use tool for Bayesian model averaging.

The rest of the paper is organized as follows: Section 2 briefly outlines Bayesian model averaging for linear regression models with the MC^3 sampling algorithm and jointness measures. Section 3 provides an overview of the **gretl** packages for Bayesian model averaging. Section 4 includes an overview of Bayesian model averaging software packages. Section 5 presents an empirical illustration. The final section concludes.

2 Bayesian inference in normal linear regression models

In this section, we briefly introduce the main features of Bayesian inference in linear regression models. We present Bayesian estimation and prediction with normal-gamma natural conjugate priors and many explanatory variables, as well as model selection and Bayesian

model averaging techniques. Finally, in this section we present the basics of the Markov chain Monte Carlo model composition sampling algorithm and jointness measures.

2.1 Bayesian estimation, model selection, and prediction

Let us consider normal linear regression models which differ in their explanatory variables (see Koop 2003 for further details). Suppose that we have K potential explanatory variables, which means there are 2^K possible models and let M_r for $r = 1, \ldots, 2^K$ denote the 2^K different models under consideration. Suppose also that y_i and x_i denote the observed data on the dependent and explanatory variables for $i = 1, \ldots, N$. The observations are placed in a $(N \times 1)$ vector y and a $(N \times k_r)$ matrix X_r containing the set of regressors included in model M_r[1]. Thus, we can write our model as

$$y = \alpha\iota_N + X_r\beta_r + \epsilon,$$

where ι_N is a $(N \times 1)$ vector of ones, β_r is a $(k_r \times 1)$ vector of unknown parameters, ϵ is a $(N \times 1)$ vector of errors which are assumed to be normally distributed, $\epsilon \sim N(0_N, h^{-1}I_N)$ and h is the error precision, which is defined as $h = \frac{1}{\sigma^2}$. $N(\mu, \Sigma)$ denotes a normal distribution with mean μ and variance Σ. Following Koop (2003), the prior for β_r is normally distributed,

$$\beta_r \mid h, M_r \sim N\left(0_{k_r}, h^{-1}\left[gX_r^\top X_r\right]^{-1}\right),$$

while we use a noninformative prior for the intercept and precision,

$$p(\alpha) \propto 1, p(h) \propto \frac{1}{h}.$$

The factor of proportionality g is part of the so-called Zellner's g-prior (Zellner, 1986). This prior is a convenient way to specify the prior variance matrix, because it reduces the number of prior variance parameters and considerably simplifies posterior computations. The **gretl** package offers the five most popular alternative Zellner's g-priors (see Fernández *et al.* 2001a and Moral-Benito 2010).

- Unit information prior (g-UIP), recommended by Kass and Wasserman (1995):

$$g = \frac{1}{N}. \tag{1}$$

- Risk inflation criterion (g-RIC), proposed by Foster and George (1994):

$$g = \frac{1}{K^2}. \tag{2}$$

- Benchmark prior, recommended by Fernández *et al.* (2001a):

$$g = \begin{cases} \frac{1}{K^2} & \text{for } N \leq K^2, \\ \frac{1}{N} & \text{for } N > K^2. \end{cases} \tag{3}$$

[1] We subtract the mean from all regressors as in Fernández *et al.* (2001a).

- g-HQ prior which mimics the Hannan and Quinn criterion, see Fernández *et al.* (2001a):

$$g = \frac{1}{(\ln N)^3}. \tag{4}$$

- Root of g-UIP, see Fernández *et al.* (2001a):

$$g = \sqrt{\frac{1}{N}}. \tag{5}$$

By Bayes' rule, the mean of the posterior distribution of the slope parameters β_r, conditional on model M_r, can be written as

$$\mathsf{E}(\beta_r \mid y, M_r) = \left[(1+g)\, X_r^\top X_r\right]^{-1} X_r^\top y. \tag{6}$$

It is easy to see that if $g \approx 0$ the mean of the posterior distribution (see Equation 6) equals the OLS estimates. The posterior variance of β_r, conditional on model M_r, is given by

$$\mathsf{VAR}\left(\beta_r \mid y, M_r\right) = \frac{N s_r^2}{N-2}\left[(1+g)\, X_r^\top X_r\right]^{-1}, \tag{7}$$

where

$$s_r^2 = \frac{\frac{1}{1+g} y^\top P_{W_r} y + \frac{g}{1+g}\left(y - \bar{y}\iota_N\right)^\top \left(y - \bar{y}\iota_N\right)}{N}$$

and $P_{W_r} = I_N - W_r\left(W_r^\top W_r\right)^{-1} W_r^\top$ for $W_r = (\iota_N, X_r)$.

The marginal data density, conditional on model M_r, may be written as

$$p(y \mid M_r) \propto \left(\frac{g}{1+g}\right)^{\frac{k_r}{2}} \left[\frac{1}{1+g} y^\top P_{W_r} y + \frac{g}{1+g}\left(y - \bar{y}\iota_N\right)^\top \left(y - \bar{y}\iota_N\right)\right]^{-\frac{N-1}{2}}. \tag{8}$$

In the Bayesian approach of comparing models, it is considered useful to employ probabilities to represent the degree of belief associated with the alternative models. For the normal linear regression models we can easily test two mutually exclusive (non-nested) and jointly exhaustive models with different subset of variables. Using Bayes' theorem, the posterior odds ratio for a model M_l against model M_n is given by

$$\frac{\mathsf{P}(M_l \mid y)}{\mathsf{P}(M_n \mid y)} = \frac{\mathsf{P}(M_l)}{\mathsf{P}(M_n)} \frac{p(y \mid M_l)}{p(y \mid M_n)}, \tag{9}$$

where $\frac{\mathsf{P}(M_l)}{\mathsf{P}(M_n)}$ is the prior odds ratio and $\frac{p(y|M_l)}{p(y|M_n)}$ is the Bayes factor. If the ratio $\frac{\mathsf{P}(M_l|y)}{\mathsf{P}(M_n|y)}$ is larger than 1, we can say that the data supports model M_l over model M_n. In our package, we use two popular model priors.

- Binomial prior:

$$\mathsf{P}(M_r) = \theta^{k_r}(1-\theta)^{K-k_r} \quad \text{for } r = 1, \ldots, 2^K. \tag{10}$$

Note that for $\theta = 0.5$ we have a uniform prior on the model space, i.e., $\mathsf{P}(M_r) = 2^{-K}$.

- Binomial-beta prior (see Gelman, Carlin, Stern, and Rubin 1997):

$$\mathsf{P}(\Xi = k_r) = \frac{\Gamma(K+1)}{\Gamma(k_r+1)\Gamma(K-k_r+1)} \cdot \frac{\Gamma(a+k_r)\Gamma(K+b-k_r)}{\Gamma(a+b+k_r)} \cdot \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}, \tag{11}$$

  where Ξ denotes model size.

In our package, we only need to specify the prior expected model size $\mathsf{E}(\Xi) \in (0, K]$. Note that in case of the binomial prior we have $\mathsf{E}(\Xi) = K\theta$ and the choice of $\mathsf{E}(\Xi)$ automatically produces a value for the prior inclusion probability θ. If we have the binomial-beta prior, the average model size will satisfy $\mathsf{E}(\Xi) = \frac{a}{a+b}K$. Here, we follow Ley and Steel (2009) and fix $a = 1$ hence obtaining the value of the second hyperparameter as $b = \frac{K-\mathsf{E}(\Xi)}{\mathsf{E}(\Xi)}$.

One of the most widely accepted norms of scientific investigation is Occam's razor. Simply speaking, Occam's razor states that the simplest hypothesis is also the most probable (see for example Madigan and Raftery 1994). In the **BMA** package we penalize large models through the prior model probabilities (Osiewalski and Steel 1993). Note that one can make $\mathsf{P}(M_r)$ a decreasing function of k_r simply by defining the prior expected model size $\mathsf{E}(\Xi)$. The smaller the prior expected model size $\mathsf{E}(\Xi)$, the less probable are large models.

It is easy to show that the posterior probability of model M_l is given by

$$\mathsf{P}(M_l \mid y) = \frac{\mathsf{P}(M_l)p(y \mid M_l)}{\sum_{r=1}^{2^K} \mathsf{P}(M_r)p(y \mid M_r)}.$$

The posterior density of vector β is the average of the posterior densities $p(\beta_r \mid y, M_r)$ conditional on the models

$$p(\beta \mid y) = \sum_{r=1}^{2^K} \mathsf{P}(M_r \mid y)p(\beta_r \mid y, M_r).$$

Once the model posterior probabilities have been calculated, we can also easily evaluate the mean and variance of the posterior distribution of slope parameters (see Leamer 1978).

$$\mathsf{E}(\beta \mid y) = \sum_{r=1}^{2^K} \mathsf{P}(M_r \mid y)\mathsf{E}(\beta_r \mid y, M_r) \tag{12}$$

and

$$\mathsf{VAR}(\beta \mid y) = \sum_{r=1}^{2^K} \mathsf{P}(M_r \mid y)\mathsf{VAR}(\beta_r \mid y, M_r) + \sum_{r=1}^{2^K} \mathsf{P}(M_r \mid y)\left(\mathsf{E}(\beta_r \mid y, M_r) - \mathsf{E}(\beta \mid y)\right)^2 \tag{13}$$

In a similar manner, we can find other characteristics of the posterior distribution (see for example Koop 2003, p. 266). We might be also interested in the estimates of the posterior inclusion probability $\mathsf{P}(i \mid y)$ (PIP), i.e., the probability that, conditional on the data, but unconditional with respect to a specific model, the variable x_i is relevant in explaining the dependent variable y (see Leamer 1978; Mitchell and Beauchamp 1988; Doppelhofer and Weeks 2009). The PIP is calculated as the sum of the posterior model probabilities for all of the models including variable x_i.

In Bayesian model averaging approach, predictions of the dependent variable are made by taking a weighted average of the individual model forecasts. As described in Fernández *et al.* (2001b), prediction of an unobserved data point y_f is based on calculating

$$p(y_f \mid y) = \sum_{r=1}^{2^K} f_S(y_f \mid N-1, \bar{y} + \frac{1}{g+1} x_{f,r}^\top \beta_r^*, \\ \frac{N-1}{N s_r^2}\{1 + \frac{1}{N} + \frac{1}{g+1} x_{f,r}^\top (X_j^\top X_j)^{-1} x_{f,r}\}^{-1}) \mathsf{P}(M_r \mid y), \quad (14)$$

where $f_S(y_f \mid v, b, a)$ denotes the univariate Student-t density with v degrees of freedom, mean b and precision a (with variance $v/\{a(v-2)\}$). In addition, $\beta_r^* = \left[X_r^\top X_r\right]^{-1} X_r^\top y$ is the ordinary least squares estimator for β_r and $x_{f,r}$ groups the r elements of x_f corresponding to the regressors in model M_r.

2.2 MC^3 sampling algorithm

Our MC^3 sampling algorithm is based on the Metropolis-Hastings algorithm, and was originally developed by Madigan *et al.* (1995). It simulates a chain of models $M^{(s)}$ for $s = 1, \ldots, S$ to find the equilibrium distribution $\mathsf{P}(M_r \mid y)$ of the posterior model probabilities. We do this as follows: We construct a candidate set of models, containing all models constructed by deleting one independent variable from $M^{(s-1)}$ or adding one independent variable to $M^{(s-1)}$, where $M^{(s-1)}$ is the previously accepted model $M^{(s-1)}$. The chain is then constructed by drawing a model M' from this candidate set and the acceptance probability has the form

$$\alpha\left(M^{(s-1)}, M'\right) = \min\left\{\frac{\mathsf{P}(M')p(y \mid M')}{\mathsf{P}(M^{(s-1)})p(y \mid M^{(s-1)})}, 1\right\}.$$

In order to assess the reliability and the convergence of the chain, we look at the Pearson's correlation between the analytical and MC^3 posterior model probabilities. Convergence is concluded to be achieved if the correlation is above 0.99 (see Fernández *et al.* 2001b and Koop 2003). Note that we measure correlation between the analytical and MC^3 posterior model probabilities *only* for the top ranked models. If the number of top ranked models is very small, this may lead to a high value of Pearson's correlation even when convergence has not been achieved.

2.3 Jointness measures

The main implementations of Bayesian model averaging are concerned with the selection of variables when model uncertainty is present. Another relevant issue which arises in this framework is to identify whether different sets of two variables x_i and x_j are substitutes, complements or neither over the model space. For that reason, Ley and Steel (2007) and Doppelhofer and Weeks (2009) define ex-post jointness measures of dependence between different sets of explanatory variables. The logarithm of the jointness statistic proposed by Ley and Steel (2007) has the form

$$J_{LS} = \ln\left[\frac{\mathsf{P}(i \cap j \mid y)}{\mathsf{P}(i \mid y) + \mathsf{P}(j \mid y) - 2\mathsf{P}(i \cap j \mid y)}\right], \quad (15)$$

Evidence	Jointness statistics
strong substitutes	$J_{LS}, J_{DW} \leq -2$
significant substitutes	$-2 < J_{LS}, J_{DW} \leq -1$
not significantly related	$-1 < J_{LS}, J_{DW} < 1$
significant complements	$1 \geq J_{LS}, J_{DW} < 2$
strong complements	$J_{LS}, J_{DW} \geq 2$

Table 1: Classification of strength of jointness measures.

where $\mathsf{P}(i \cap j \mid y)$ represents the sum of the posterior probabilities of those models that contain both variables x_i and x_j, $\mathsf{P}(i \mid y)$ and $\mathsf{P}(j \mid y)$ are the PIPs of x_i and x_j. J_{LS} can be interpreted as the posterior odds ratio of the models including both i and j vs the models that include them only individually (see Ley and Steel 2007).

An alternative jointness measure was proposed by Doppelhofer and Weeks (2009). It can be written as follows

$$J_{DW} = \ln \left[\frac{\mathsf{P}(i \cap j \mid y)\mathsf{P}(\tilde{i} \cap \tilde{j} \mid y)}{\mathsf{P}(i \cap \tilde{j} \mid y)\mathsf{P}(\tilde{i} \cap j \mid y)} \right], \tag{16}$$

where $\mathsf{P}(\tilde{i} \cap \tilde{j} \mid y)$ denotes the sum of the posterior probabilities of the regression models in which neither x_i and x_j are included, $\mathsf{P}(i \cap \tilde{j} \mid y)$ corresponds to the sum of the posterior probabilities of all models in which x_i is included and x_j is excluded. The last probability $\mathsf{P}(\tilde{i} \cap j \mid y)$ is defined analogously.

J_{DW} corresponds to the posterior odds of including i, given that j is included, divided by the posterior odds of including i given that j is not included (see Doppelhofer and Weeks 2009). According to Doppelhofer and Weeks (2009), we use the classification of jointness among the variables given in Table 1.

3 Implementation in gretl

In this section, we describe the code as well as the use of the **gretl** package for Bayesian model averaging, together with the accompanying jointness measures. First, we will characterize our code and the use of the GUI, then we will present how to use **BMA** via a script. Finally, we will show the outputs that are returned.

3.1 The core of the BMA code

The **BMA** package consists of 3 public and 22 private functions, but only 16 of them are used regularly. The name of each function starts with the prefix `BMA_`.

3.1.1 Public functions

The public functions are:

1. `BMA_GUI`
2. `BMA_Print`
3. `BMA_OldStyle`

The core package function which runs and controls the main loop is:

```
function bundle BMA_GUI (series Y "Dependent variable",
  list X_list "List of indenpendent variables (X)",
  int acc_type[1:2:1] "Model prior" {"Binomial", "Binomial-Beta"},
  scalar av_model_size[0::1] "Prior average model size",
  scalar alpha[0:1:0.6] "Significance level for the initial model",
  int l_rank[2::4] "Number of the top ranked models",
  int g_type[1:5:1] "g-prior type" {"Benchmark prior",
    "Unit Information Prior (g-UIP)", "Risk Inflation Criterion (g-RIC)",
    "Hannan and Quinn HQC", "Root of g-UIP"},
  int do_joint[0:2:0] "Jointness analysis" {"None", "Ley-Steel Measure",
    "Doppelhofer-Weeks Measure"},
  int h_predict[0::0] "Number of out-of-sample forecasts",
  int Nrep[1e+06] "Total number of replications",
  int burn[0:99:10] "Percentage of burn-in draws",
  int verbosity[1:2:1] "Verbosity",
  bool show_progress[1] "Show overall progress",
  bool force_dof_lack[0] "Force more variables than observations")
```

The words in quotation marks are the labels for the GUI shown in Figure 3.

- The `series Y` is just a series of dependent variable.
- The `X_list` is a **gretl**'s object "named list" which is just a set of K indenpendent variables (defined by names or dataset IDs). What is very important the `X_list` cannot contain a `const` (**gretl**'s internal and automatically generated constant term).
- The `acc_type[1:2:1]` is an integer indicating model prior type (the default is binomial, see Equations 10 and 11).
- The `scalar av_model_size[0::1]` is the scalar of the prior average model size. Note, if `av_model_size` $= \frac{K}{2}$ and the model prior is set to binomial, we get the uniform prior on the model space. The default value is 1.
- The `scalar alpha[0:1:0.6]` is the significance level in OLS estimation. An independent variable enters the initial model if its p value is less than the significance level (see Koop 2003). The default value is $\alpha = 0.6$, but setting $\alpha = 1$ results in a model consisting of 0 to K randomly chosen explanatory variables.
- The `int l_rank[2::10]` is the number of the top ranked models. The default value is 10.
- The `int g_type[1:5:1]` indicates the type of g-prior to be used (the default value is 1: "Benchmark prior"), see Equations 1, 2, 3, 4 and 5.
- The `int do_joint[0:2:0]` indicates whether we do jointness analysis and if so, which measure to use. The default value is 0: "None", see Section 2.3.
- The `h_predict[0::]` is the number of out-of-sample forecasts.

- The `int Nrep[1000000]` is the total number of replications in the Monte Carlo simulation. The default value is 1000000.
- The `int burn[0:99:10]` is the percentage of burn-in draws ranging from 0% to 99%. The default value is 10%.
- The `int verbosity[1:2:1]` indicates the level of verbosity of the **BMA** package for printing results. The default value is 1, i.e., silent mode.
- The `bool show_progress[1]` switches progress printing during the simulation. Note that it consumes a lot of CPU time and significantly slows the package.
- The `bool force_dof_lack[0]` makes the BMA analysis to run even if there is not sufficient degrees of freedom.

Function for results printing is:

```
function void BMA_Print (bundle *MODEL)
```

This function takes as an argument reference to the bundle MODEL with BMA results and prints results according to `verbosity` settings.

The last public function is `BMA_OldStyle` which is written for backwards compatibility and can be used with scripts prepared for old versions of BMA package (prior to 2.0).

3.1.2 The main loop

The main loop of the **BMA** package is split into five parts controlled by following functions:

1. `BMA_parse` – basic parsing and dataset checks,
2. `BMA_setup` – bundle setup,
3. `BMA_main` – MCMC,
4. `BMA_summary` – final computations and bundle clean up,
5. `BMA_Print` – results printing.

Essentials of BMA analysis is the MCMC stuff driven by `BMA_main`. This part is split into thee parts:

1. Setting up the MC^3 sampling algorithm.
2. Constructing the initial model.
3. MC^3 simulation.

3.1.3 The bundle returned by the BMA package

The **BMA** package can optionally return bundle containing results obtained in the analysis. The most interesting are:

- `Post_Results` containing posterior moments (unconditional and conditional on inclusion),
- `jointness_m` containing posterior joint probability of variables,
- `prediction` containing predictive results.

The structure of the `Post_Results` matrix is shown in Figure 1. It has K rows (one for each explanatory variable) and five columns are: the posterior inclusion probabilities, the posterior mean and standard deviation of each coefficient (Mean and Std.Dev.) and the posterior mean and standard deviation of each coefficient conditional on the variable being included in the model (Cond.Mean and Cond.Std.Dev.), see page 14 for details.

	PIP	Mean	Std.Dev.	Cond.Mean	Cond.Std.Dev.
x_1					
x_2					
x_3					
$\vdots$					
x_K					

Figure 1: Structure of the `Post_Results` matrix returned by the **BMA** package.

The structure of the `jointness_m` matrix is shown in Figure 2.

	x_1	x_2	x_3	...	x_K
x_1					
x_2					
x_3	joint probability of variables				
$\vdots$					
x_K					

Figure 2: Structure of the `jointness_m` matrix returned by the **BMA** package.

3.2 Usage of the BMA package

3.2.1 The GUI way

Once you start **gretl**, you must open a data file and then you can load the relevant **BMA** package from the **gretl** server. In the main window, go to Tools > Function packages > On server heading. By selecting **BMA**, you will open a window similar to the one shown in Figure 3:

According to Figure 3, we can specify the following entries in the GUI **BMA** window

- **Dependent variable** – The dependent variable.

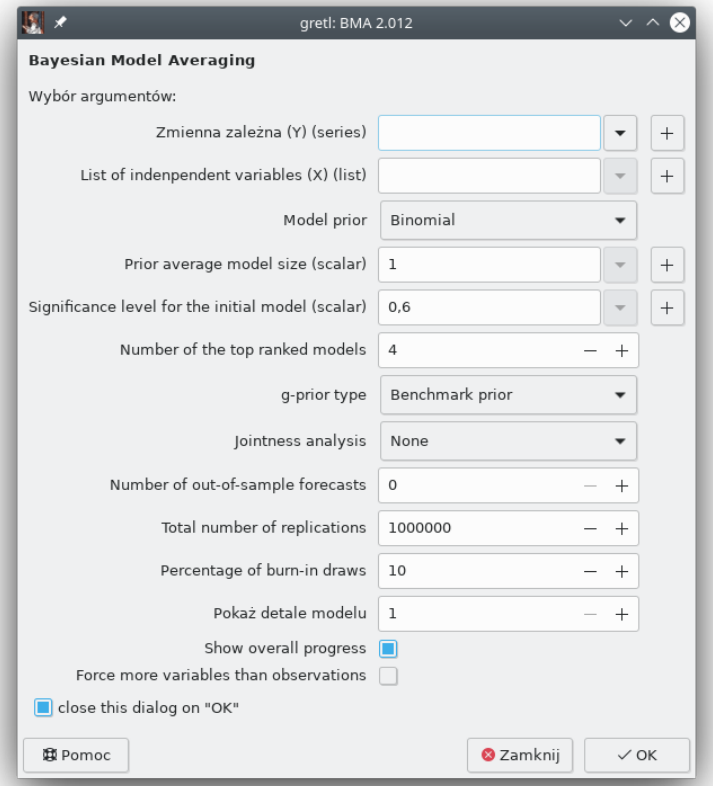


Figure 3: Main window for **BMA**.

- **List of indenpendent variables** – Loading variables from the database, which must have been opened previously. Note that by default we assume that you want to estimate an intercept; therefore, a constant is implicitly included in the list of the variables.

- **Model prior** – Indicates the choice of model prior. One can employ the binomial model prior or the binomial-beta model prior. Note that the uniform model prior is a special case of the binomial model prior. Therefore, in fact, our package allows for three types of priors.

- **Prior average model size** – Specifies the prior expected model size $\mathsf{E}(\Xi)$. The expected model size may range from 0 to K. The default value equals 1. It is the smallest reasonable prior expected model size. It means that we penalize large models and we assign high probability to small ones following Occam's razor philosophy.

  Note that for the binomial model prior and with $\mathsf{E}(\Xi) = 0.5K$ one can define the uniform prior on the model space. For example, for $K = 10$ regressors we can define the uniform prior with $\mathsf{E}(\Xi) = 5$. For $\mathsf{E}(\Xi) < 5$ we assign high probability to small models. The smaller the prior expected model size $\mathsf{E}(\Xi)$, the less probable are larger models.

- **Significance level for the initial model** – Defines the significance level which is used to

build the initial model. An explanatory variable enters the initial model if its p value is less than the significance level. If the significance level equals 1, the initial model will be randomly chosen (with equal probability) from all available models. Note that if all available explanatory variables enter the initial model, you will get the following **gretl** error message "No independent variables were omitted".

- **Number of top ranked models** – Specifies the number of best models for which detailed information is stored.
- **g-prior type** – One can choose between four Zellner's g-priors for the regression coefficients. Choices include: benchmark prior, unit information prior, risk inflation criterion, Hannan and Quinn prior, root of g-UIP.
- **Jointness analysis** – If `"None"` (the default), the jointness analysis is omitted. Alternatively, one can choose the jointness measures of Ley and Steel (2007) or Doppelhofer and Weeks (2009).
- **Number of out-of-sample forecasts** – Defines the total number of out-of-sample forecasts of the dependent variable.
- **Total number of replications** – Defines the total number of iteration draws to be sampled.
- **Percentage of burn-in draws** – Specifies the number of burn-in replications, calculated as the percentage of the total number of iteration draws.
- **Verbosity** – An integer value of 1 or 2; the default is 1, which allows to see the basic Bayesian model averaging results. If Verbosity equals 2, a more detailed description of the analysis is provided (initial model, speed of convergence, estimation results for top ranked models).
- **Show overall progress** – The output is flushed every 10% of MC^3 and the progress is printed. Note: this consumes a lot of CPU time.

3.2.2 Script

The **BMA** package can also be used inside Hansl scripts. A very minimal code could be as follows:

```
open greene9_1.gdt
include BMA.gfn
set seed 1000000
list greene = capital labor nfirm

BMA_GUI(valadd, greene, 1, 1.5, 0.6, 4, 1, 0, 0, 10000, 10, 1)
```

The above example consists of three blocks.

The first block defines just the opening of the so-called `greene9_1` dataset, which is bundled in every standard **gretl** installation. This dataset contains the cross-sectional data on the manufacturing of transportation equipment presented in Greene (1999)[2] and it consists

[2]This dataset is also available at link `http://people.stern.nyu.edu/wgreene/Text/Edition7/TableF7-2.txt`.

of the following variables: valadd – output, capital – capital input, labor – labor input and nfirm – number of firms.

The second block is the definition of the `greene` list which contains indenpendent variables available in the `greene9_1` dataset.

The third block contains the specification of the Bayesian model averaging analysis: binomial model prior, prior average model size set to 1.5 (which means that we have the uniform model prior), significance level for the initial model set to 0.6, 4 top ranked models, Benchmark g-prior, without jointness analysis, without out-of-sample forecasts, 1000000 replications with 10% burn-in draws basic output (verbosity set to 1) and progress overall progress is printed..

Suppose we want to run the same analysis but following Occam's razor philosophy. The simplest way to do this, is to set the prior average model size to 1. The code would be as follows:

```
BMA_GUI(valadd, greene, 1, 1, 0.6, 4, 1, 0, 0, 10000, 10, 1, 1)
```

In this case, we penalize large models by assigning high probabilities to small ones. Apart from the binomial model prior, we can use the binomial-beta prior. To do this, we have to set the model prior to 2. The code would be as follows

```
BMA_GUI(valadd, greene, 2, 1, 0.6, 4, 1, 0, 0, 10000, 10, 1, 1)
```

Suppose we want to set the g-prior to risk inflation criterion, select the uniform model prior, do jointness analysis with the Ley-Steel measure, compute forecasts for 3 observations and print additional information (verbosity set to 2). The code would then be as follows:

```
BMA_GUI(valadd, greene, 1, 1.5, 0.6, 4, 3, 1, 3, 10000, 10, 2, 0)
```

Finally, if we want to save results of the above Bayesian model averaging analysis in the the bundle `results_b`, the code should be as follows

```
results_b = BMA_GUI(valadd, greene, 1, 1.5, 0.6, 4, 3, 1, 3, 10000, 10, 2, 0)
```

Suppose we want to compute Doppelhofer-Weeks Measure instead of already computed Ley-Steel Measure. Since the type of jointness measure is controlled by 'do_joint' field in the result bundle, the only thing we have to do is to set the value of 'do_join' to 2 and print BMA results again. The code should be as follows:

```
results_b.do_joint = 2
BMA_Print(&results_b)
```

If we need backwards compatibility with BMA_package prior to version 2 we can use the BMA_OldStyle function to run the old style script.

```
list greene_old = valadd capital labor nfirm
BMA_OldStyle(greene_old, 1, 1.5, 0.6, 4, 1, 1, 3, 10000, 10, 2)
```

3.2.3 Outputs

If you select the appropriate entries in the GUI **BMA** window, our package returns: PIP, Mean, Std.Dev., Cond.Mean, Cond.Std.Dev. Furthermore, the package also returns the predictive results of the dependent variable (Mean and Std.Dev.).

Let us consider the data used in Fernández *et al.* (2001b) (denoted FLS hereafter). This data comprises the information about 72 countries and 41 potential growth determinants for the period 1960 to 1992[3]. The initial estimation sample consists of indices $i = 1, \ldots, 69$ and the out-of-sample evaluation indices are from 70 to 72.

For example, for the FLS data, the following estimates should appear

```
Posterior moments (unconditional and conditional on inclusion):
                 PIP         Mean     Std.Dev.    Cond.Mean   Cond.Std.Dev
   GDPsh560   0.998378    -1.444579     0.305391    -1.446927       0.300031
Confuncious   0.890878     0.427081     0.189631     0.479393       0.123638
   Life_Exp   0.932297     0.988063     0.404005     1.059816       0.314688
  Equip_Inv   0.904971     0.549653     0.250217     0.607370       0.184734
  SubSahara   0.654270    -0.431560     0.369153    -0.659606       0.240550
     Muslim   0.363769     0.130498     0.197138     0.358739       0.157977
Rule_of_Law   0.416318     0.206002     0.271416     0.494818       0.184491
   Yrs_Open   0.547074     0.312026     0.318952     0.570354       0.196508
        ...        ...          ...          ...          ...            ...
----------------------------------
Posterior probability of models:
Model 1:    0.007899
Model 2:    0.005294
Model 3:    0.004204
Model 4:    0.003960
Model 5:    0.003378
     ...         ...
Total probability of the models in ranking (numerical): 0.058997
Correlation coefficient between the analytical
and numerical probabilities of the above models: 0.997518
----------------------------------
Predictive results:
            Growth       Mean   Std.Dev.
Obs.70    2.714600   1.590657   1.279354
Obs.71    2.067600   2.006087   0.941517
Obs.72    1.889700   2.465582   0.981258
```

The **BMA** estimate function accepts a scalar, which sets the verbosity of the output. Its default value is 1, which causes the estimation output to be printed out. A value of 2 forces the **BMA** function to print out all the details of the estimation. You can print out the above-mentioned results and additionally the following information: the total CPU time, type of

[3]The dataset is publicly available on the *Journal of Applied Econometrics* online data archive (`http://qed.econ.queensu.ca/jae/2001-v16.5/fernandez-ley-steel/`). The reported chain (3000000 replications with 33% burn-in draws) took about 51 minutes of CPU time on a PC with AMD Phenom II X6 1100T CPU, 6.0 Gb of RAM running under Debian GNU/Linux. We used **gretl** 1.10.2 compiled by GCC 5.2.1. The seed for RNG was set to 1000000. The dependent variable is the growth rate from 1960–1992.

model prior, prior average model size, significance level for the initial model, type of g-prior, total number of iterations and, finally, the number of burn-in draws. Moreover, the **BMA** estimate function produces the estimation results for the initial and top ranked models. The jointness analysis is inactive by default. If it is active, you will get: posterior joint probability of explanatory variables, jointness statistic as given in Equations 15 or 16 and classification of jointness measures. The jointness analysis for the previous example should look like this:

```
Posterior joint probability of variables:
             GDPsh560 Confuncious   Life_Exp  Equip_Inv ...
   GDPsh560   0.000000    0.889256   0.932182   0.903349 ...
Confuncious   0.000000    0.000000   0.826208   0.816956 ...
   Life_Exp   0.000000    0.000000   0.000000   0.839068 ...
  Equip_Inv   0.000000    0.000000   0.000000   0.000000 ...
        ...        ...         ...        ...        ... ...
Jointness statistics (Ley-Steel Measure):
             GDPsh560 Confuncious   Life_Exp  Equip_Inv ...
   GDPsh560   0.000000    2.083161   2.643181   2.235000 ...
Confuncious   0.000000    0.000000   1.576599   1.618373 ...
   Life_Exp   0.000000    0.000000   0.000000   1.662559 ...
  Equip_Inv   0.000000    0.000000   0.000000   0.000000 ...
        ...        ...         ...        ...        ... ...
Strong substitutes:
        Rev_Coup,Area       -4.516140
Rev_Coup,Publ_Edu_pct       -4.459739
                  ...             ...
Significant substitutes:
       GDPsh560,Hindu       -1.997989
     Mining,War_Dummy       -1.989541
                  ...             ...
Significant complements:
 Confuncious,Life_Exp        1.576599
Confuncious,Equip_Inv        1.618373
                  ...             ...
Strong complements:
  DPsh560,Confuncious        2.083161
   GDPsh560,Equip_Inv        2.235000
                  ...             ...
```

4 Comparison to other BMA software packages

Several Bayesian model averaging software packages have been developed by many researchers[4]. The most popular in the literature are R packages. A recent and comprehensive overview of Bayesian model averaging software in R is given in Amini and Parmeter (2011). They outline the following BMA packages: **BMS** by Zeugner and Feldkircher (2015), **BAS** by Clyde,

[4]Useful information about Bayesian model averaging software packages are available at link `http://bms.zeugner.eu/resources/`.

Main features	R-**BMS**	R-**BMA**	R-**BAS**	**FLSBMA**	**gretl-BMA**
Model sampling	MC^3	BIC	BAS, AMCMC, BAS+MCMC	MC^3	MC^3
Model priors	5	1	3	2	3
Zellner's g-priors	6	1	9	9	5
Jointness analysis				✓	✓
Jointness measures				1	2
Out-of-sample forecast	✓		✓	✓	✓
No. of burn-in draws	✓		✓	✓	✓
No. of draws	✓	✓	✓	✓	✓
No. of top ranked models	✓			✓	✓
Graphical user interface					✓

Table 2: Main features of the selected Bayesian model averaging software packages. Model sampling: Type of model sampling algorithms. Model priors: Number of available model priors. Zellner's g-priors: Number of available Zellner's factors of proportionality. Jointness analysis: Information if the package provides jointness analysis. Jointness measures: Number of available jointness measures. Out-of-sample forecast: Information about which package can be used for out-of-sample forecasting. No. of burn-in draws. No. of draws and no. of top ranked models: Number of user-defined parameters in the sampling algorithm. Graphical user interface: Presence of a GUI in the package.

Littman, and Ghosh (2012) and **BMA** by Raftery, Hoeting, Volinsky, Painter, and Yeung (2014). According to their research the **BMS** package provides the most accurate results and it is also the most flexible with numerous options. Table 2 presents the main features of the above mentioned packages. Furthermore we compare our **BMA** package to the **FLSBMA** program developed in Fortran 77 by Fernández *et al.* (2001b).

The most flexible package according to the model sampling category is R-**BAS** which contains three sampling algorithms: Bayesian adaptive sampling (BAS), adaptive MCMC (AMCMC) and Bayesian adaptive sampling with MCMC for the initial replications (BAS+MCMC). All packages offer different model priors. The most limited is R-**BMA** with only a fixed model prior, while the most comprehensive is R-**BMS** which contains the following model priors: fixed, uniform, binomial, binomial-beta and custom (user-controlled prior probabilities). Furthermore, the number of g-priors may differ depending on the package starting from one for R-**BMA**, to nine for R-**BAS** and **FLSBMA**.

gretl-BMA and **FLSBMA** are the only packages which offer Bayesian model averaging together with accompanying jointness statistics. **FLSBMA** contains one jointness measure (Ley-Steel) while **gretl-BMA** contains two measures (Ley-Steel and Doppelhofer-Weeks). The estimation of out-of-sample forecasts is possible in almost all packages excluding R-**BMA**, where this option is not available. Three out of five packages enable full control of the replication algorithm. In R-**BMS**, **FLSBMA** and **gretl-BMA**, the user can set the number of burn-in draws, replication draws and detailed information about top ranked models. Unlike other packages **gretl-BMA** offers an easy-to-use GUI for the Bayesian model averaging approach.

In order to compare run times of the above mentioned packages we used the FLS dataset with the following options: benchmark g-prior, 1000000 burn-in draws, 2000000 iterations. Model prior was set to uniform. We considered two scenarios for our experiment: with or

Run time	R-**BMS**	R-**BMA**	R-**BAS**	**FLSBMA**	**gretl-BMA**
Without jointness analysis	317	92	53	1423	1852
With jointness analysis				1423	3073

Table 3: Run time in seconds of Bayesian model averaging software packages for FLS data.

without jointness analysis.

According to the results presented in Table 3, the fastest Bayesian model averaging software package is R-**BAS**, while the slowest one is **gretl-BMA**. We are aware of this disadvantage of our package, but it strongly depends on the **gretl** development process, especially on the improvements of the Hansl interpreter. It is possible that in the near future **gretl** will be much faster than the current 1.10.2 version (see Section 6 for more details).

5 Empirical illustration

In this section we examine the ability of our package in replicating the results published by Fernández *et al.* (2001b) and Ley and Steel (2007). We use the same original dataset to attempt to replicate their results. In our empirical illustration, we discard the first 1 million models and draw samples from the model space 2 million times. We specify the following entries in the GUI **BMA** window: model prior = 'binomial', prior average model size = '20.5' (we set the model prior to the uniform distribution), number of the top ranked models = '20', g-prior type = 'Benchmark prior', total number of replications = '3000000', percentage of burn-in draws = '33'. Tables 4 and 5 present the estimation results[5]. They report the posterior means, standard errors and PIPs of regressors calculated by R-**BMS**, R-**BMA**, R-**BAS** and **FLSBMA** packages. These benchmarking results allow us to compare and analyze the performance of our package. Bold font indicates identical results.

As is apparent in Tables 4 and 5, the only minor differences in the posterior results are found between **gretl-BMA** and the results published in Ley and Steel (2007). All PIPs and estimated posterior means or standard deviations are reasonably close in all cases and the same variables are identified to be relevant. Note that the **gretl** package results are almost identical to the results produced by the R-**BMS** package.

In our opinion, the main reason for the very similar results produced by R-**BMS** and **gretl-BMA** is the implementation of the same RNG algorithm (the SIMD-oriented Fast Mersenne Twister), while **FLSBMA** package uses an older RNG algorithm, i.e., "ran2" developed by L'Ecuyer. The R-**BMA** and R-**BAS** packages are not capable to reproduce our results (see Amini and Parmeter 2011 for additional information).

6 Conclusions

This paper has outlined a new software package that implements Bayesian model averaging and jointness measures for **gretl**. Bayesian model averaging is a straightforward and natural extension of standard Bayesian analysis and it is a useful and popular alternative to other

[5]The reported chain took about 31 minutes of CPU time on a PC with AMD Phenom II X6 1100T CPU, 6.0 Gb of RAM running under Debian GNU/Linux. We used **gretl** 1.10.2 compiled by GCC 5.2.1. The seed for RNG was set to 1000000.

	R-**BMS**		R-**BMA**		R-**BAS**		**FLSBMA**		**gretl-BMA**	
Regressors	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Log GDP in 1960	−1.425	0.280	−1.595	0.195	−1.467	0.241	−1.418	0.267	−1.427	0.275
Fraction Confucian	**0.494**	0.127	0.641	0.089	0.579	0.103	0.490	0.117	**0.494**	0.126
Life expectancy	0.954	0.396	1.024	0.213	1.004	0.242	0.958	0.370	0.967	0.386
Equipment investment	0.553	0.237	0.510	0.121	0.455	0.164	0.557	0.222	0.550	0.234
Sub-Saharan dummy	−0.472	0.346	−0.919	0.172	−0.848	0.214	−0.476	0.334	−0.474	0.346
Fraction Muslim	0.260	0.228	0.149	0.156	0.141	0.198	0.257	0.219	0.261	0.229
...					...					
Population growth	**0.005**	**0.048**	−0.004	0.037	0.006	0.060	0.003	0.035	**0.005**	**0.048**
British colony dummy	**−0.003**	**0.031**	0.360	0.133	0.159	0.196	−0.002	0.022	**−0.003**	**0.031**
Outward orientation	−0.003	0.029	−0.160	0.099	−0.074	0.102	−0.002	0.021	−0.004	0.030
Fraction Jewish	**−0.002**	**0.027**	-0.001	0.010	0.000	0.022	−0.001	0.020	**−0.002**	**0.027**
Revolutions and coups	**0.000**	**0.023**	n/a	n/a	**0.000**	0.024	**0.000**	0.017	**0.000**	**0.023**
Public education share	**0.001**	**0.023**	0.014	0.053	0.061	0.109	0.000	0.017	**0.001**	**0.023**
Area (scale effect)	**−0.001**	**0.021**	n/a	n/a	−0.003	0.030	**−0.001**	0.014	**−0.001**	**0.021**

Table 4: The posterior estimates of regression coefficients based on the Bayesian model averaging software packages for the FLS data. The dependent variable is the growth rate from 1960–1992 across 72 countries. All the regressors have been centered to have zero mean. Abbreviation "n/a" means that R-**BMA** dropped variable in pre-selection procedure. Bold font denotes identical results.

Regressors	R-**BMS**	R-**BMA**	R-**BAS**	**FLSBMA**	**gretl-BMA**
Log GDP in 1960	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
Fraction Confucian	**0.99**	1.00	1.00	1.00	**0.99**
Life expectancy	0.93	1.00	1.00	0.95	0.94
Equipment investment	**0.92**	1.00	0.97	0.94	**0.92**
Sub-Saharan dummy	**0.74**	1.00	1.00	0.76	**0.74**
Fraction Muslim	**0.65**	0.60	0.43	0.66	**0.65**
...			...		
Population growth	**0.04**	**0.04**	0.09	0.02	**0.04**
British colony dummy	**0.04**	0.98	0.49	0.02	**0.04**
Outward orientation	**0.04**	0.86	0.44	0.02	**0.04**
Fraction Jewish	**0.04**	0.01	0.06	0.02	**0.04**
Revolutions and coups	**0.03**	n/a	0.06	0.02	**0.03**
Public education share	**0.03**	0.10	0.32	0.02	**0.03**
Area (scale effect)	**0.03**	n/a	0.06	0.02	**0.03**

Table 5: Posterior inclusion probabilities based on the Bayesian model averaging software packages for FLS data. Abbreviation "n/a" means that R-**BMA** dropped the variable in the pre-selection procedure. Bold font denotes identical results.

variable selection procedures, especially for a large set of regressors. Here we used **gretl**, which is a free, open-source software for econometric analysis with an easy-to-use GUI. Our goal was to familiarize potential users with the features and the different options that our package has to offer. We described how our package implements Bayesian model averaging, as well as the outputs that are returned.

Recent trends in Bayesian model averaging still concern applications in various areas of empirical research, i.e., economic growth (see Crespo Cuaresma, Doppelhofer, and Feldkircher 2014, Gazda and Puziak 2012), stock returns (see Beckmann and Schüssler 2014), geography (see Baran 2014) and many more. One can also encounter new papers about theory of Bayesian model averaging, for example: two-stage Bayesian model averaging (see Lenkoski, Eicher, and Raftery 2014) or comparison of Bayesian model averaging to the weighted average least squares method (see Luca and Magnus 2011).

The future development of the **BMA** package could be focused on two main aspects: speed improvements and feature extensions. The first one is strongly correlated with the **gretl** development process, especially with work on improvements in speed of the Hansl interpreter. It is possible that the interpreter will be rewritten in the near future to achieve speed boost with backward compatibility. The feature extensions would concern two aspects: implementing new g-priors and model priors. In the literature we can find many types of g-priors (see Fernández *et al.* 2001a). Moreover new model priors could be also considered. For instance, Sala-I-Martin, Doppelhofer, and Miller (2004) proposed the so-called fixed model prior. These implementations would surely improve the flexibility of the **BMA** package.

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